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Bench-Scale Liquid-Phase Microaeration and Iron Dosing Comparison for In Situ Biogas Desulfurization and Antibiotic Resistance Management in Mesophilic Anaerobic Digestion

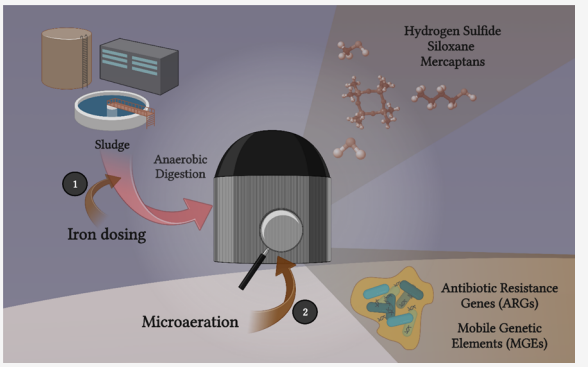
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ABSTRACT: Microaeration and iron dosing are promising in situ desulfurization methods for anaerobic digestion (AD). However, limited reports exist on their systematic comparison. Consequently, this study investigated liquid-phase microaeration (60−106 mL air/$L_{reactor}$/d) and ferric chloride (250−650 mg $FeCl_3$/L) dosages in mesophilic AD of sewage sludge operated at a solid residence time of 20 days. Both methods provided up to 88% H_2S removal and moderate siloxane removal (30−37%) but had no apparent impact on mercaptan removal from biogas. However, the control and test reactors demonstrated equivalent AD performance in terms of biomethane yields, chemical oxygen demand, and total/volatile solids degradation efficiencies, indicating that air and iron dosages were primarily utilized for biogas desulfurization. Still, both methods resulted in substantial changes in microbial community composition. Notably, the indication of syntrophy between facultative microbes and hydrogenotrophic methanogens was intensified by both methods. For both methods, antibiotic resistance gene (ARG) and mobile genetic element (MGE) abundances in digestate decreased considerably due to the decrease in the abundance of potential ARG hosts. Compared to iron dosing, MGE removal was slightly higher in the microaeration reactor. These findings provide valuable insights into the efficacy of these two approaches in desulfurization and the reduction of antibiotic resistance in AD.



KEYWORDS: *anaerobic digestion, sludge, in situ biogas desulfurization, siloxanes, mercaptans, microaeration, iron dosing, antibiotic resistance genes*

INTRODUCTION

In most centralized water resource recovery facilities (WRRFs) worldwide, mesophilic or thermophilic anaerobic digestion (AD) is widely practiced for sludge stabilization.[1,2] Biogas from AD can be used to co-generate heat and electricity or can be upgraded to renewable natural gas (RNG). However, extensive purification requirements of biogas to remove hydrogen sulfide (H_2S), mercaptans, and volatile organic compounds (e.g., siloxanes) limit its conversion to heat/electricity/RNG, resulting in ubiquitous biogas flaring instead.[3] Particularly, excessive H_2S and mercaptans in the biogas can cause corrosion of metal pipes, combustion engines, and biogas utilization equipment.[3] Moreover, silicon dioxide particles produced from the combustion of siloxanes can cause extensive damage to process equipment by erosion or blockage.[4] The release of sulfur oxides (SO_x) in flue gases is increasingly becoming subject to air quality monitoring and permit restrictions. Thus, regardless of biogas end use, reducing sulfides allows WRRFs to maintain high operational and safety standards and better odor control.

Established biogas desulfurization methods include various end-of-the-pipe, ex situ approaches, such as scrubbing, biofilter, biotrickling filters, absorption, and adsorption, where the biogas is cleaned ex situ in a separate downstream treatment unit.[5] Despite being commercially available and implemented to full-scale, ex situ methods (e.g., absorption) are usually considered expensive (US $0.244−2.38/$m^3$ biogas).[5] Thus, researchers have explored the use of in situ approaches, such as microaeration and iron salt addition, for biogas desulfurization. Microaeration is operated via controlled injection of a small amount of air or pure oxygen into the headspace or liquid phase of digesters.[6] At low oxygen levels, sulfide can be biochemically oxidized to elemental sulfur and/or sulfate. Iron

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(salts/oxides) dosing is another approach to achieving in situ desulfurization.[5] Liquid-phase microaeraion can be operated through various methods, such as injecting directly into the digester, in the sludge feeding process, or through a recirculation line. Liquid-phase microaeration is often recommended for its ease of operation and control and has been shown to be effective in full-scale applications.[5] Divalent (Fe^{2+}) and trivalent (Fe^{3+}) iron salts are commonly used and can react with sulfide to form iron sulfide precipitates.[5,6] As summarized in a recent review report,[6] a few full-scale demonstrations have already been reported for both methods, showing effective H_2S removal (72−99%) performance. Although most studies focused on mesophilic AD, studies suggested that both methods could be effective for mesophilic and thermophilic AD. Compared to ex situ desulfurization technologies, microaeration and iron dosing have emerged as potential low-cost alternatives (~US $0.01−0.016/$m^3$ biogas).[5] These methods are mostly localized within the digester and limit dissolved sulfide conversion to H_2S.[5] However, microaeration is considered relatively underdeveloped due to its commercial unavailability, lack of estimated costs associated with the maintenance of digesters in case of elemental sulfur deposition, and safety concerns around dosing the incorrect amount of oxygen/air.[6] Although both methods have been widely explored as low-cost alternatives for biogas desulfurization, there is limited information on their systematic comparison for biogas desulfurization and efficacy in removing other contaminants such as mercaptans and siloxanes from biogas.

Iron dosing for desulfurization is often considered as a relatively more established method than microaeration, as it has been widely used for sulfide removal in sewer systems for many years.[7] However, the use of microaeration for in situ biogas desulfurization has recently received attention as it is considered the chemical-free treatment alternative, potentially resulting in less cost.[5,6] Nonetheless, a recent economic analysis report based on data collected from different studies suggested that the operating costs of the two methods would be competitive, e.g., US $0.01/$m^3$ for iron dosing versus US $0.016/$m^3$ for microaeration.[5] The water sector would benefit from further systematic comparisons using performance data collected under identical test conditions to provide a more reliable decision support tool in process selection.

Wastewater treatment plants (WWTPs) are considered as hotspots for antibiotic resistance gene (ARG) propagation and transmission to other ecosystems.[8] High concentrations of residual antibiotics and pathogenic microorganisms in WWTPs are stimulating the resistance prevalence among microbial communities.[8,9] ARGs can be disseminated among microbial communities by horizontal gene transfer (HGT) via mobile genetic elements (MGEs) such as plasmids and integrons (i.e., *intl*1 and *intl*2) that are considered as biomarkers for HGT.[10,11] A few recent reports also suggested that both methods (microaeration and iron dosing) could improve the removal of ARGs and MGEs in AD and limit their transmission to our ecosystem to a certain extent.[12,13] The deaths related to antibiotic-resistant infections have rapidly increased globally in recent years, raising concerns regarding potential ARG/MGE spread routes.[14] As the land application of digestate presents a major route of ARG/MGE spread,[15] enhancing their removal along with desulfurization would be an added benefit and a step toward achieving the circular economy and environmental sustainability. As of now, no study on a systematic comparison of the effectiveness of these two methods in removing ARGs/MGEs has been reported in the literature.

To fill the aforementioned research gaps, this study presents a systematic, bench-scale comparison of liquid-phase microaeration and iron dosing for in situ biogas desulfurization in mesophilic AD of sewage sludge. First, the effects of two methods on H_2S, mercaptan, and siloxane removals were investigated along with AD performance. Moreover, variations of ARG/MGE abundances and microbial communities were analyzed. To the best of the authors' knowledge, in the past, no such comprehensive report was available for the comparison of these two methods under identical experimental conditions.

MATERIALS AND METHODS

Experimental Set-Up. The experiments were carried out using six identical glass anaerobic bioreactors (ISES-Canada, Vaughan, ON, Canada) with working and total volumes of 1.5 and 2 L, respectively. The bioreactors are equipped with mechanical agitators coupled with electrical motors and reactor lids with ports that can be used for either gas collection or injection (Figure S1). Bioreactors were operated in a water bath (VWR Water bath 28L, 120V, VWR Canada, Mississauga, ON, Canada) to maintain a mesophilic operating condition at 37 ± 1 °C. The liquids were stirred with agitators at 200 rpm. Duplicate bioreactors were operated for each condition, and average results are reported here.

Sludge Characteristics. The digested sludge (inoculum) and AD feed sludge used for the experiments were collected from a wastewater treatment plant in Edmonton (Alberta, Canada). The digested sludge was collected once during the start-up of the reactors, while the AD feedstock sludge was collected monthly. Before use, all samples were stored at 4 °C in a cold room, for a period of 1−4 weeks, depending on the sampling frequency. Before the start-up of the reactors, the inoculum was pre-incubated for a few days at 37 °C for acclimation of microbial communities and residual organics degradation. The average characteristics of digested sludge were as follows: total chemical oxygen demand (TCOD): 23 ± 0.1 g/L, soluble chemical oxygen demand (SCOD): 3.5 ± 0.1 g/L, total solids (TS): 29 ± 1 g/L, volatile solids (VS): 24 ± 2 g/L, alkalinity: 9.5 ± 0.6 g/L as $CaCO_3$, and pH: 7.1 ± 0.1. The average feed sludge characteristics were as follows: TCOD: 70 ± 4 g/L, SCOD: 7.1 ± 0.4 g/L, TS: 43 ± 2 g/L, VS: 36 ± 2 g/L, dissolved sulfide (S^{2-}): 35 ± 1 mg/L, and pH: 6.9 ± 0.2.

Start-Up, Operation, and Experimental Phases. During start-up, 750 mL of inoculum and 750 mL of AD feed sludge (substrate) were added to each digester. Thus, the initial ratio of feed to inoculum was set at 1.5 (VS basis), following recommended ratios of ≤2 found in the literature,[16,17] to prevent excessive organic loading in the digesters. The bioreactors were purged with nitrogen gas for 5 min to ensure anaerobic conditions. For the first 3 days, reactors were operated in a batch mode, and then the operation was switched to a semicontinuous mode. A solid residence time (SRT) of 20 days was selected for the experiment to mimic AD conditions at the wastewater treatment plant. Based on the working volume of the reactor, the sludge was fed to the bioreactor at a rate of 75 mL/d using peristaltic pumps (Longer Pump BT100−2 J, Langer Instruments Corp, Tucson, AZ), which corresponds to the organic loading rate (OLR) of 1.47 kg COD/m^3/d. Before starting experiments with microaeration and iron dosing, all six reactors

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Table 1. Summary of Experimental Phases

experimental phases	control	microaeration	iron dosing
pre-experimental phase (days 1−34)			
phase-1 (days 35−46)		60 mL air/d (40 mL air/$L_{reactor}$/d)	250 mg $FeCl_3$/L
phase-2 (days 47−94)		120 mL air/d (80 mL air/$L_{reactor}$/d)	500 mg $FeCl_3$/L
phase-3 (days 95−125)		160 mL air/d (106 mL air/$L_{reactor}$/d)	650 mg $FeCl_3$/L

were operated under similar operating conditions to establish a baseline for subsequent comparison. This operational phase was referred to as the 'pre-experimental phase'. It should be noted that SRT (20 days) remained constant throughout the operation of 125 days. Based on the observed methane yields and the removal of organics (discussed later), it was evident that the dosages of iron and microaeration employed during the entire operation remained low, thereby having minimal impact on methane productivity and kinetics. The primary objective of the pre-experimental phase was to establish that biogas yield and quality were consistent across all reactors, thus providing a baseline for commencing microaeration and iron dosing experiments. This approach enables a reliable comparison of different conditions (i.e., control vs microaeration vs iron dosing) regarding their effects on AD performance. After ensuring comparable performance (based on methane gas production and H_2S content) from all six reactors, microaeration and iron dosing experiments were started in duplicate. Four reactors were used for microaeration and iron dosing experiments, and two reactors were operated as a control (without microaeration or iron dosing). For microaeration operation, the air was injected directly into the liquid phase through the gas injection port located in the lids of the reactors using a peristaltic pump. However, the pump was intermittently operated every 4 h for a specific time to maintain the target daily air supply rate (Table 1). The total pump run time for the microaeration operation in phase-1, phase-2, and phase-3 were 5, 10, and 15 min per day, respectively. The selected microaeration conditions (40−106 mL air/$L_{reactor}$/d) were within the range of effective microaeration conditions, providing >80% H_2S removal, reported in the literature (3−160 mL air/$L_{reactor}$/d).[18−21] It should be noted that some studies also used extremely high microaeration with air, which could cause the dilution of biogas. For instance, Yang and Deng[22] studied microaeration by injecting air at 66.7 mL/L/min in a 4.5 L digester (i.e., 21,344 mL air/$L_{reactor}$/d). Among different forms of iron (di- and trivalent salts/oxides) used for in situ desulfurization, Fe (III) is known to provide faster reaction kinetics and is also effective for a wide pH range.[23,24] Therefore, ferric chloride ($FeCl_3$) was selected for this testing. For iron dosing, feed sludge was amended with $FeCl_3$ (iron(III) chloride, reagent grade, 97%, Sigma Aldrich, Canada) before daily feeding. As summarized in Table 1, three different dosages of microaeration and $FeCl_3$ have been tested in this study.

Analytical Methods. TCOD, SCOD, and alkalinity were measured using the HACH reagent kits (HACH Co., Loveland, CO). For SCOD analysis, samples were filtered using 0.45 μm membrane filters. For COD analysis, HACH COD standard solution (1000 mg/L as COD) and deionized water were used as the standard and blank, respectively. The total solid (TS) and volatile solid (VS) concentrations were measured using standard methods (2540 Solids).[25] pH was measured using a bench-top pH meter (Accumet AR15, Fisher Scientific, Pittsburgh, PA). The oxygen reduction potential (ORP) was measured with a portable ORP meter (Milwaukee MW500, ORP Meter with platinum electrode, ±1000 mV, 1 mV resolution, ±5 mV accuracy, Milwaukee Instruments, Rocky Mount, NC). Dissolved sulfide (S^{2-}) was measured using the Hach Sulfide Reagent Set, methylene blue (#2244500, Hach Canada, London, ON). To avoid high sulfide concentrations which can inhibit the full-color development, samples were diluted (10 times).

Biogas produced from the bioreactors was collected in gas bags, and then volumes were measured using frictionless glass syringes. Biogas composition (CH_4, H_2, and CO_2 content) was analyzed using gas chromatography (7890B, Agilent Technologies, Santa Clara) equipped with a thermal conductivity detector (TCD) and two columns (Molsieve 5A 60/80 mesh and Hayesep N 80−100 mesh). All methane production data were expressed at standard temperature and pressure (STP) (0 °C, 1 atm). Gaseous H_2S concentrations were measured with a portable H_2S gas monitor (Acrulog H_2S Short Deployment Gas Monitor 0−2000 ppm, Model ACRSD2000, Detection Instruments Corp., Phoenix, AZ). After each measurement, the sampling tubes were purged with clean air before the following measurement. Siloxane and mercaptan concentrations in biogas were analyzed by ALS Environmental (Waterloo, ON, Canada). Siloxane analysis was performed using procedures adapted from EPA Method TO-17, ISO Method 16017, and NIOSH Method 2549. Collected biogas samples actively collected on thermal desorption sorbent tubes were thermally stripped, and the analytes were recollected on the trapping material of a focusing trap in the thermal desorber. The analytes were then thermally desorbed into a thermal desorption−gas chromatography−mass spectrometry (TD-GC-MS) system for analysis. Mercaptan concentrations were analyzed by gas chromatography with a sulfur chemiluminescent detector (SCD). The analysis was performed using procedures adapted from ASTM D5504. Biogas samples are collected into cleaned evacuated silica-coated canisters. By means of a loop system, a volume of biogas is transferred from the canister and cryofocused before determining the mercaptan compounds by GC-SCD.

Microbial Community and DNA Extraction. After the completion of phase-3, digested sludge samples were aseptically collected from each treatment for the bacterial and archaeal community analysis. After centrifugation at 15 min at 2000×g, the pellet (0.25 g) was taken for the genomic DNA extraction using the PowerSoil DNA Isolation Kit (MoBio Laboratories, Carlsbad) as per the manufacturer's instructions. The quality and yield of the extracted DNA were detected by a spectrophotometer (NanoDrop 2000C, Thermo Scientific). The genomic DNA was sent to the Research and Testing Laboratory (Lubbock, TX) for Illumina Miseq sequencing of the 16S rRNA gene using the universal primer set 515F/806R.[10] For the microbial community and diversity, the raw sequencing data were processed using the Quantitative Insights Into Microbial Ecology (QIIME) pipeline (QIIME2, Version 2021.2). The detailed methodology can be found

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elsewhere.[26] The chord diagram was generated similar to our previous study[27] by using operational taxonomic units (OTUs) that were analyzed using the QIIME2 software.

Quantification of ARGs. Tetracycline resistance genes (*tet*M, *tet*Q, *tet*X, *tet*W, *tet*B, and *tet*A), sulfonamide resistance genes (*sul*1 and *sul*2), β-lactam resistance genes (bla_{TEM}), and macrolide resistance genes (*erm*C and *erm*B) were evaluated by the quantitative polymerase chain reaction (qPCR). Similarly, MGEs including integrons (*intl*1 and *intl*2) were quantified by qPCR. The detailed quantification procedures were performed according to the QuantiFast SYBR Green PCR Kit's protocol using the CFX 96 real-time PCR system with a C1000 Thermal Cycler (Bio-Rad). The detailed methodology and primers can be found elsewhere.[8]

Data Analysis. Student's *t*-test was conducted using Microsoft Excel to identify statistically significant differences in the results based on average values at a 95% confidence level. To analyze the relationship between ARG gene copies and the relative abundance of the microbial community, correlation analysis was conducted using Spearman's Rho coefficient. The correlation was visualized using a correlation plot, which was created using the R software (RStudio v. 1.4.1103). The correlation was represented by a range of values from −1 to +1 and *p*-values less than 0.05 were considered to be statistically significant.

RESULTS AND DISCUSSION

Anaerobic Digestion Performance. Figure 1 shows the daily methane production from bioreactors under different operating conditions over 125 days of operation. Between days 1−34, all reactors were operated under identical conditions, i.e., the pre-experimental phase without microaeration and iron dosing, to establish a baseline for subsequent comparison. During the first week of operation, methane production profiles in all reactors showed considerable fluctuations. However, the daily methane production rate slightly decreased and stabilized during the following week. This stable daily methane production continued slightly over a week. After day 17, the daily methane production slightly increased in all reactors. However, between days 20−34, average daily methane production in control, microaeration, and iron-dosed reactors were comparable (0.22 ± 0.05, 0.19 ± 0.08, 0.21 ± 0.06 L $CH_4/L_{reactor}$/d, respectively). Furthermore, methane content (vol.%) in biogas produced from these reactors was quite similar (50 ± 2, 51 ± 5, and 52 ± 5%, respectively) (*p*-value = 0.235) (see Figure S2). The H_2S content in biogas measured between days 25−28 was also comparable (*p*-value = 0.606) in all reactors (see Figure S2). Thus, despite some fluctuations, all reactors showed quite similar performance in terms of methane productivity and biogas quality (i.e., CH_4 and H_2S content). Therefore, this state was considered a baseline for starting microaeration and iron dosing experiments, allowing a reliable comparison among different conditions (i.e., control vs microaeration vs iron dosing).

On day 35, phase-1 operation of microaeration (40 mL air/$L_{reactor}$/d) and iron (250 mg $FeCl_3$/L) dosing was started and was continued until day 46 (Figure 1). During that period, methane production and methane content in biogas increased in all reactors and reached a pseudo-steady-state. The average daily methane production in the control, microaeration, and iron-dosed reactors were 0.39 ± 0.05, 0.41 ± 0.02, and 0.44 ± 0.06 L $CH_4/L_{reactor}$/d, respectively. The methane contents

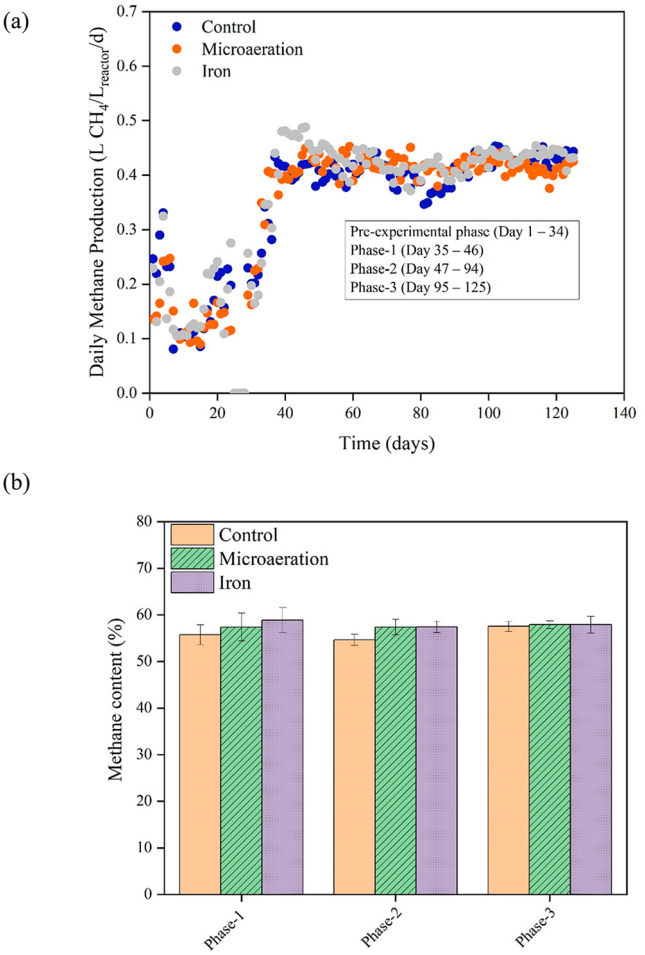


Figure 1. (a) Daily methane production and (b) average methane content (%) in biogas during different experimental phases (data are presented as the average ± standard deviation (number of samples from duplicate reactors, *n* = 8, 72, and 60 for phase-1, -2, and -3, respectively)).

(Figure 1b) in biogas were 56 ± 2% (control), 57 ± 3% (microaeration), and 59 ± 3% (iron dosing). Thus, methane production and methane content in biogas were similar for all conditions. As discussed later, only H_2S content in biogas was appreciably impacted by microaeration and iron dosing. Thus, air and iron dosages selected for phase-1 were mostly utilized for desulfurization without impacting the methanogenic activity. If these technologies were to affect methanogenic activity adversely, they would not be appealing to the management of AD facilities.

Consequently, to examine the impact of increased microaeration and iron dosing, air and iron dosages were increased during phases 2 and 3. Compared to phase-1, air and iron dosages were doubled during phase-2 (days 47−94). However, analogous to phase-1, the methane production rate and methane content remained unaffected (Figure 1). Daily methane production in the control, microaeration, and iron-dosed reactors were 0.40 ± 0.02, 0.42 ± 0.02, and 0.42 ± 0.02 L $CH_4/L_{reactor}$/d, respectively. The methane contents in biogas were 55 ± 1, 57 ± 2, and 57 ± 1%, respectively (Figure 1). Thus, daily methane production and methane content in biogas were comparable between the two phases (phase-1 and -2) despite significant differences in air (40 vs 80 mL air/$L_{reactor}$/d) and iron (250 vs 500 mg $FeCl_3$/L) dosages. Despite

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a further increase in air and iron dosages to 106 mL air/$L_{reactor}$/d and 650 mg $FeCl_3$/L, respectively, in phase-3, the methane production rate and methane content in biogas remained the same.

Figure S3 illustrates effluent TCOD, SCOD, TS, and VS concentrations under different operating conditions. Under each phase, effluent TCOD, SCOD, TS, and VS concentrations were comparable in all reactors. Thus, these data also corroborate the methane production profiles. Considering the theoretical methane yield of 350 mL CH_4/g $COD_{removed}$ at STP, the COD mass balances computed based on experimental methane production and TCOD removal indicated a closure of 90 to 120% (see Figure S3), thus emphasizing data reliability.[28]

As discussed later, H_2S content in biogas in microaeration and iron-dosed reactors gradually decreased between phase-1 and phase-3 (discussed later). Thus, microaeration and iron dosing appeared to have a trivial impact on methane production and were primarily utilized for biogas desulfurization. To date, a few studies investigated liquid-phase microaeration.[18,21,22,29] Jeníček et al.[21] and Yang and Deng[22] reported that microaeration could increase methane production by 17% and 6.4−19.6%, respectively. Notably, Yang and Deng[22] suggested that methane production increased due to enhanced hydrolysis triggered by microaeration. However, the authors used considerably higher air dosing (66.7 mL/L/min in a 4.5 L digester) than our study, while H_2S in biogas was lower than in our reactors (336 ppm vs ∼550−900 ppm). It is expected that the differences in feedstock type, OLRs, SRTs, microaeration dosages, and sulfide levels may lead to different results. For instance, Nghiem et al.[29] reported no impact on methane production due to microaeration. Two previous reports based on full-scale demonstrations of iron (ferrous and ferric) also suggested a minimal effect on methane production while considerably reducing H_2S content in biogas.[30] Thus, the trivial impact of microaeration and $FeCl_3$ on methane production observed in our study was consistent with the literature.

A few reports previously raised concerns that microaeration with air instead of pure oxygen would cause the dilution of biogas with nitrogen gas,[31,32] which would reduce methane content. Although we did not measure nitrogen content in biogas, the average biogas methane content from the microaeration reactor was slightly higher than the control during all experimental phases (see Figure 1b). Thus, there was no sign of decreased methane content in biogas. Jeníček et al.[21] also previously demonstrated that microaeration with air would not considerably reduce methane content in biogas in full-scale anaerobic digesters; the maximum decrease in methane concentration was <2%. Interestingly, using pure oxygen instead of air has often been recommended by previous studies (mostly related to gas-phase microaeration) to avoid the dilution of biogas.[32] However, our results further suggested no requirement of replacing air with pure oxygen during liquid-phase microaeration, which can often be expensive.[21] It should be noted that combining oxygen with biogas is typically viewed unfavorably due to the risk of explosions linked to methane/oxygen mixtures. Despite the amount of pure oxygen dosed for microaeration being very small and may be quickly consumed, it is crucial to acknowledge that the risk of explosion is always present when working with pure oxygen for microaeration in AD, and should not be underestimated, as suggested in the literature.[6,32] Therefore, from an operational standpoint, AD facilities generally prefer not to work with pure oxygen for safety reasons, if possible. This makes air injection a more practical and cost-effective solution.

H_2S Removal from Biogas. Figure 2a shows H_2S content in biogas during different experimental phases. During the pre-

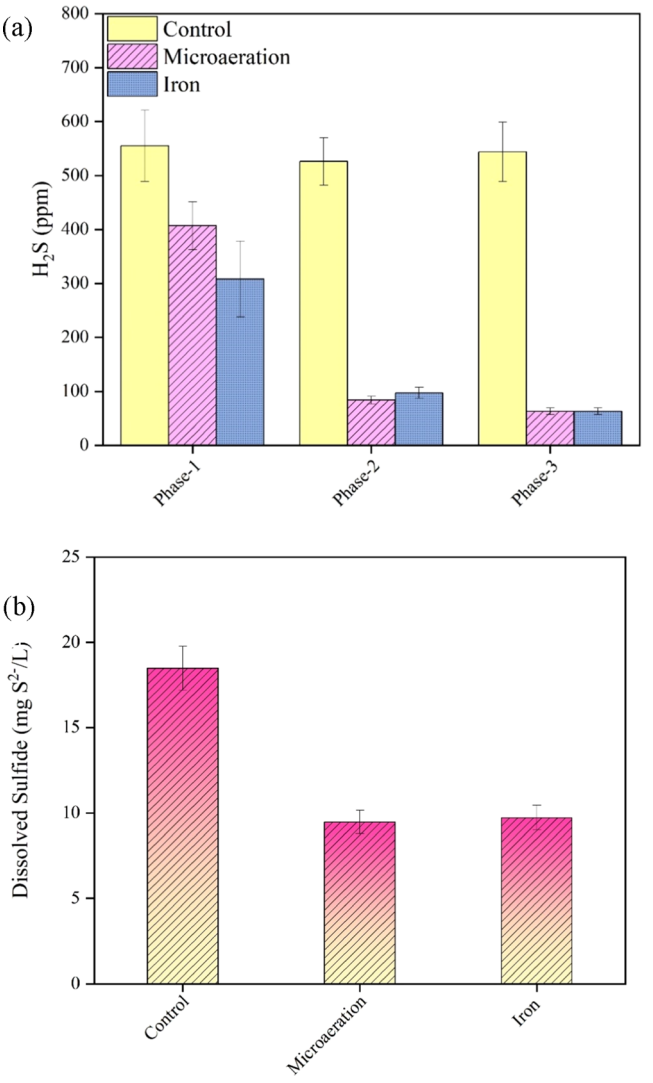


Figure 2. (a) Average H_2S content in biogas during the experimental phases (number of samples from duplicate reactors, n = 20, 72, and 60 for phase-1, -2, and -3, respectively) and (b) effluent dissolved sulfide concentrations in phase-3 (n = 32). Data are presented as the average ± standard deviation.

experimental phase, baseline H_2S content in the control, microaeration, and $FeCl_3$ reactor were 956 ± 173, 849 ± 51, and 879 ± 271 ppm, respectively. In phase-1, H_2S content in the control, microaeration, and $FeCl_3$ reactors were 555 ± 66, 407 ± 44, and 308 ± 70 ppm, respectively (Figure 2a). Compared to the pre-experimental phase, H_2S concentration in the control digester considerably decreased (p-value = 0.039) during phase-1. During the pre-experimental phase, H_2S content in biogas was measured between days 25−28. The reactors did not reach a steady state at that time. Furthermore, such changes could be attributed to different factors, including shifting microbial communities toward the steady state and slight changes in sludge characteristics (e.g., dissolved sulfide, proteins, etc.). Nonetheless, compared to the H_2S level in the

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control (555 ± 66 ppm), microaeration (60 mL air/d) and $FeCl_3$ dosing (250 mg/L) could remove H_2S in biogas by 27 and 45%, respectively, in phase-1. During phase-2, H_2S content in biogas further decreased in microaeration and $FeCl_3$ dosed reactors (110 ± 49 and 132 ± 92 ppm, respectively) while remaining almost constant in the control (549 ± 38 ppm) (Figure 2a). Thus, the average H_2S removed observed for microaeration and $FeCl_3$ were 80 and 76%, respectively. However, with further increase in microaeration and $FeCl_3$ dosages to 106 mL air/$L_{reactor}$/d and 650 mg $FeCl_3$/L, respectively, phase-3 provided moderate improvement in H_2S removals (88%) as compared to phase-2. Nonetheless, it was apparent that increasing air and $FeCl_3$ dosing could increase desulfurization efficiencies.

The maximum H_2S removal efficiencies observed for liquid-phase microaeration and $FeCl_3$ dosing (up to 88%) was within the range of high H_2S removal efficiencies previously reported in the literature (liquid-phase microaeration: 70−99%; $FeCl_3$ dosing: 72−99%).[6,23] However, in phase-1, microaeration provided considerably lower desulfurization efficiencies than $FeCl_3$ (27 vs 45%). The process of microaeration integrates both chemical and biological reactions for H_2S removal.[5,33] Particularly, the enrichment of sulfide-oxidizing bacteria (SOB) was reported as the key factor in microaerobic desulfurization.[32] It is expected that enriching the sulfide-oxidizing bacterial community would require some time. For instance, a previous study suggested that a full-scale micro-aerobic anaerobic digester would require 3−12 weeks to reach a stable desulfurization performance.[21] Jensen et al.[34] reported SOB median growth rates of 1.25 and 1.33 d^{-1} as determined from the utilization rates of hydrogen sulfide and oxygen. Iron salts (e.g., $FeCl_3$) can react with dissolved sulfide to form insoluble iron sulfide via chemical precipitation, consequently preventing the stripping of H_2S into the biogas.[5] Based on a full-scale demonstration, Erdirencelebi and Kucukhemek[30] reported that the digesters stabilized within 11 days after the $FeCl_3$ was started. Thus, it can be assumed that the response of the digester to iron dosing would be relatively faster than microaeration. Nonetheless, during phase-2, microaeration could provide slightly higher average H_2S removal than $FeCl_3$ (76 vs 80%), and comparable performance in phase-3. Analogous to H_2S levels in biogas, dissolved sulfide (S^{2-}) concentrations were comparable in microaeration (9.5 ± 1 mg/L) and iron-dosed (9.7 ± 1 mg/L) reactors in phase-3 (see Figure 2b). In contrast, dissolved sulfide concentration in the control was relatively higher (18 ± 1 mg/L). Considering the initial dissolved sulfide level of 35 ± 1 mg/L, dissolved sulfide removal efficiencies (72−73%) were lower than H_2S removal efficiencies in biogas. Previously, Krayzelova et al.[35] reported that dissolved sulfide removal was only 20−30%, while H_2S removal in biogas was as high as 79% with liquid-phase microaeration. Such an observation can be explained by the fact that H_2S oxidation also takes place in the headspace, even when air is injected into the liquid phase.[35] Furthermore, the degradation of sulfur-containing proteins (mostly sludge-bound proteins) can also produce additional sulfides and mercaptans. It should be noted that concentrations of dissolved sulfide and H_2S in the digester may change with pH.[5] Therefore, digester pH was continuously monitored (data not shown). Digester pH remained almost stable (at ∼7.5) throughout the operating period in all reactors. Thus, it was confirmed that the differences in biogas H_2S levels were not attributed to any pH changes.

Mercaptans and Siloxane Levels in Biogas. Figure 3a shows a heatmap showing the concentration profiles of various

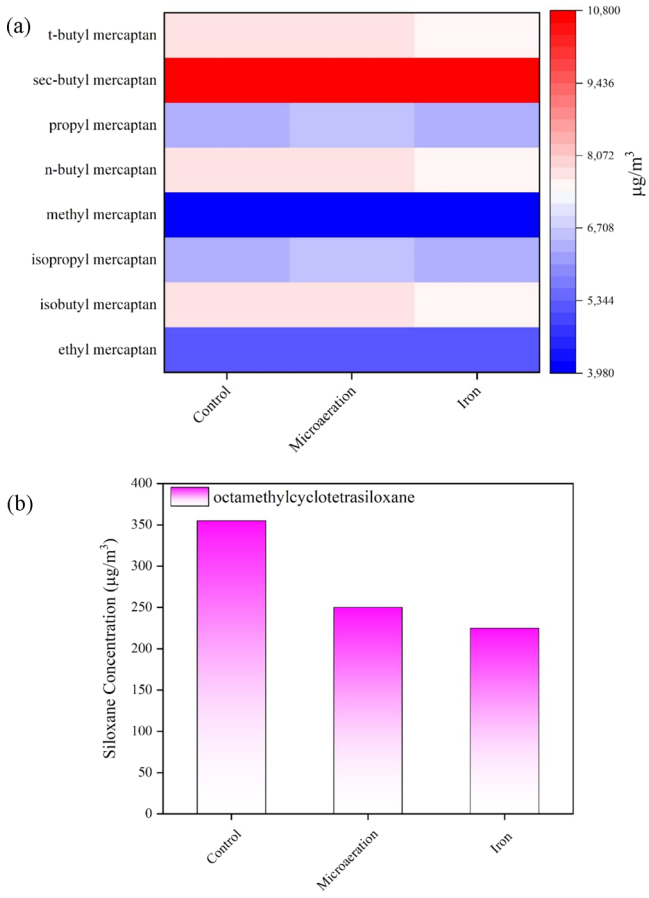


Figure 3. (a) Heatmap showing concentrations of various mercaptan compounds and (b) siloxane (D4) levels in biogas during phase-3. Data are presented as the average of two samples.

mercaptans detected in biogas. Sec-butyl mercaptans had the highest concentrations in all samples, followed by *t*-butyl mercaptan, n-butyl mercaptan, and isobutyl mercaptan compounds. Compared to the control, iron dosing provided minor removal of various mercaptans (except for sec-butyl mercaptan). However, in all cases, the differences between the control and test reactors were ≤170 $\mu g/m^3$. On the other hand, except for sec-butyl mercaptan, levels of other mercaptans in a microaeration reactor increased by 45−90 $\mu g/m^3$ more than the control. Thus, both methods appeared to have a trivial impact on mercaptan concentrations in biogas. Mercaptans are mostly generated via the degradation of sulfur-containing proteins (mostly sludge-bound proteins); the process can also produce additional H_2S.[36] Furthermore, the biochemical transformation between various mercaptan compounds may lead to changes in different mercaptan levels.[36]

Figure 3b shows the concentrations of octamethylcyclotetrasiloxane (D4), the most common siloxane compound found in AD biogas.[37,38] Iron dosing was slightly more effective than microaeration (37 vs 30% removal). Iron dosing in digesters can provide polymerization of cyclic siloxanes, which can remain in the sludge effluent.[39] Also, it has been previously reported that facultative microbes under microaerobic conditions in a biotrickling filter could remove siloxane from

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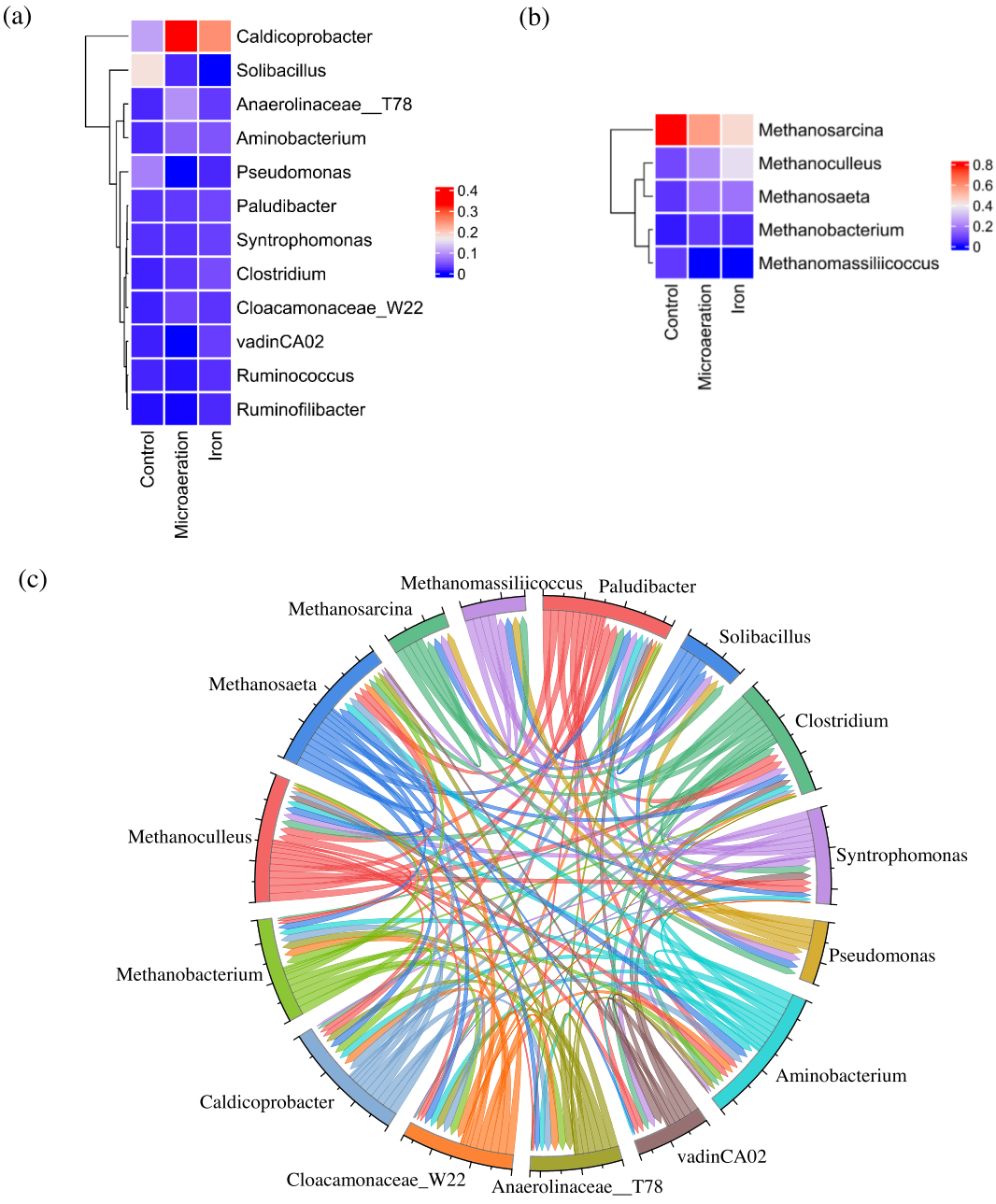


Figure 4. Microbial relative abundance at (a) the bacterial genus, (b) the archaeal genus level in phase-3, and (c) the chord diagram of the bacterial and archaeal communities.

biogas.[38] Slightly inferior siloxane removal in microaeration might be attributed to the high dissociation energy of the siloxane bonds (Si−O and Si−C).[37] However, both methods in this study provided moderate siloxane D4 removal efficiencies of <40%. Overall, both methods provided high H_2S removal efficiencies but were relatively ineffective or moderately effective for mercaptans and siloxane removal. Thus, further downstream biogas purification may still be required depending on the target end use of biogas.

Microbial Diversity and Composition. The α diversity metrics were computed to compare the microbial community diversity between the reactors (Table S1). The microbial richness of the Chao1 index and observed species (OTUs) showed the highest values in the control reactor (174 and 181) compared to the microaeration (150 and 160) and iron-dosed (128 and 132) reactors, respectively. Similar patterns were observed in the community evenness of the control reactor (0.92) compared to the microaeration (0.84) and iron salt (0.83) reactors. Compared to the control reactor (0.91), a lower Simpson reciprocal index was observed in the microaeration (0.87) and iron-dosed (0.85) reactors, suggesting lower community evenness in these digesters. The microbial diversity in terms of the Shannon index was also higher in the control (6.8), with similar values in the other reactors (6.2 and 6.3). These results demonstrated the higher richness and diversity of the microbial community in the control reactor compared to other reactors. These results suggest that microaeration and iron dosing limited the enrichment of specific microbial species while proliferating others.

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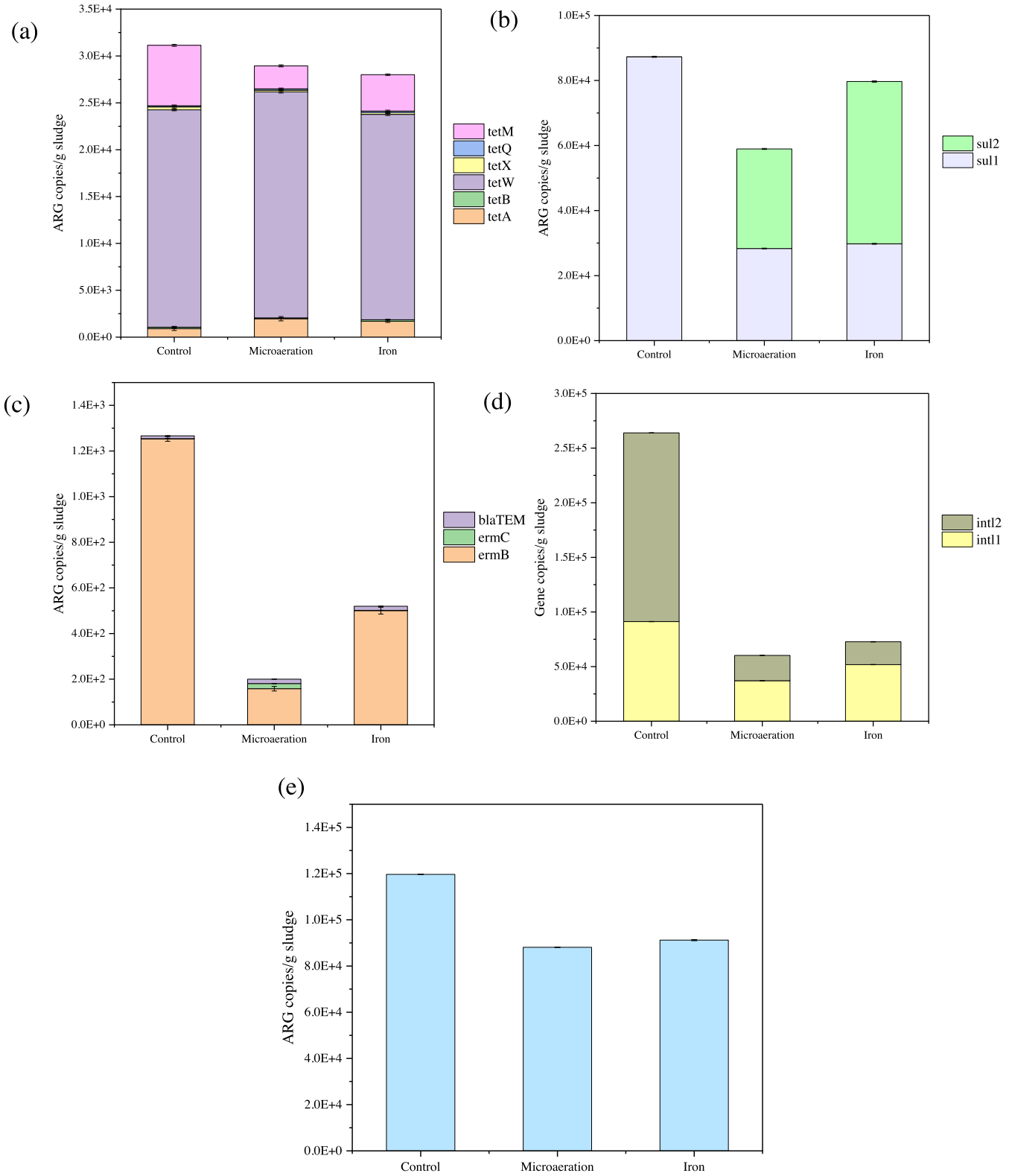


Figure 5. (a) Tetracycline resistance genes, (b) sulfonamide resistance genes, (c) macrolide and β-lactam resistance genes, (d) integrons, and (e) total ARGs in digested sludge samples from phase-3.

Based on the phylum level, *Firmicutes* was the most dominant phylum in all reactors with a higher abundance in microaeration (46%) and iron-dosed reactors (43%) than the control (37%) (see Figure S4). Members of *Proteobacteria* (26%) were the second most abundant phylum in the control reactor, followed by *Actinobacteria* (9%) and *Bacteroidetes* (7%). However, for microaeration and iron dosing, *Proteobacteria* (9 and 8%), *Synergistetes* (8 and 10%), and *Bacteroidetes* (5 and 7%) were similarly enriched, respectively. *Chloroflexi* was only enriched in the microaeration reactor (9%) with <2% in the iron-dosed and control reactors. As shown in Figure 4a, in the control reactor, *Solibacillus* was the highest abundant

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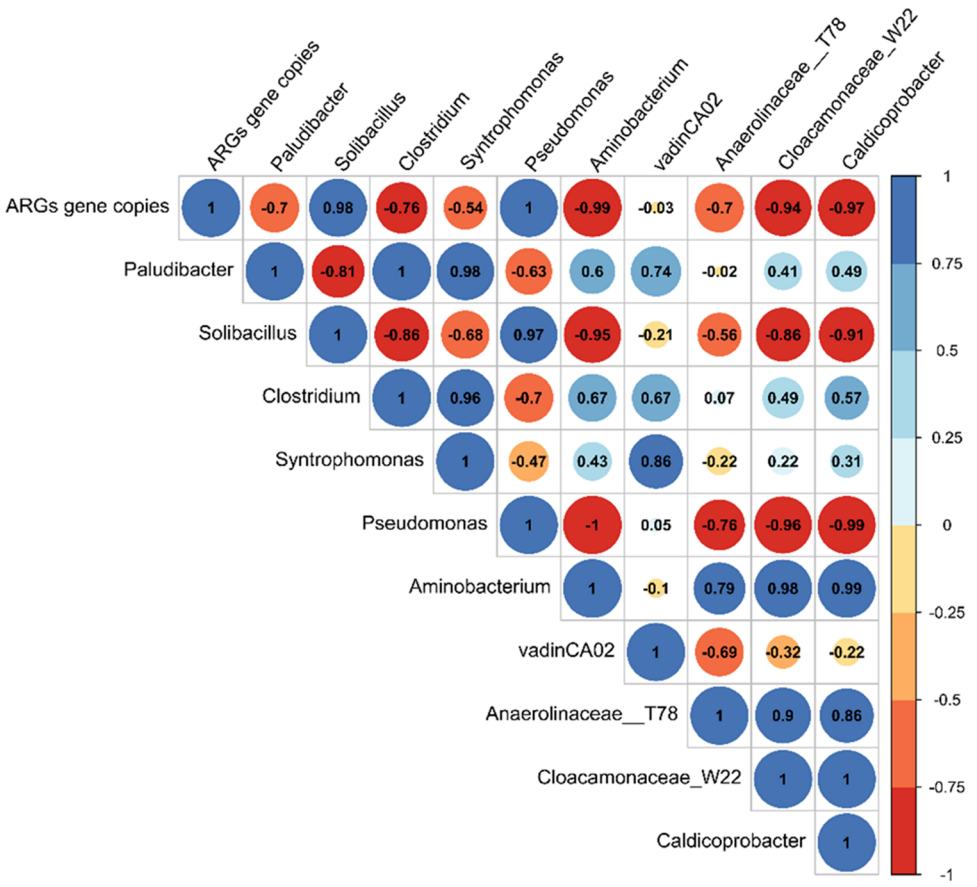


Figure 6. Spearman correlation analysis between ARG abundance and microbial communities depicting rho values (all of the *p*-values are significant).

genus with 17%, followed by *Caldicoprobacter* (10%) and *Pseudomonas* (8%). For microaeration, *Caldicoprobacter* within the class *Clostridia* was the most dominant genus (32%), followed by *Anaerolinaceae*_T78 (9%) and *Aminobacterium* (5%). Similarly, in the iron-dosed reactor, *Caldicoprobacter* and *Aminobacterium* were also prevalent with a relative abundance of 25 and 5%, respectively, with the enrichment of *Clostiridum* (4%).

As shown in Figure S4, methanogenic *Euryarchaeota* phyla enriched in higher abundances in the microaeration (18%) and iron-dosed (19%) reactors than in the control (14%). Several archaeal genera enriched in different reactors included *Methanobacterium, Methanoculleus, Methanosaeta, Methanosarcina*, and *Methanomassiliicoccus* (Figure 4b). *Methanosarcina* was the most abundant genus in all reactors, with the highest in the control (77%) reactor, followed by microaeration (56%) and iron-dosed (43%) reactors. However, for microaeration and iron dosing, several other archaeal genera were highly enriched *Methanoculleus* (21 and 36%) and *Methanosaeta* (16 and 17%), respectively. For control, *Methanoculleus* (9%) and *Methanosaeta* (6%) also existed but with less abundance than the test reactors. Instead, *Methanomassiliicoccus* was observed in the control reactor (6%) with no abundance in other reactors.

It was apparent that microaeration and iron dosing exerted selective pressures on the microbial communities, but methane production remained the same in all reactors. Thus, the observed differences in microbial communities indicate that both methods mainly altered biochemical pathways. The key difference observed between the control and test reactors was the changes in the abundance of genera *Caldicoprobacter* (25−32% in test reactors vs 10% in the control). *Caldicoprobacter* sp. can ferment a wide range of organics,[40] but they are sensitive to high sulfate levels.[41] Sulfate can be reduced to sulfide during AD. Thus, the decrease in dissolved sulfide levels in test reactors might increase their abundance in test reactors. The abundances of amino acids degrading *Aminobacterium*[42] were slightly higher in the microaeration (5%) and iron-dosed (5%) reactors than in the control (2%), showing a sign of higher protein metabolism, which might be associated with ineffective removal of mercaptans. Both microaeration and iron dosing can stimulate the growth of facultative microbes.[22,43] A syntrophic partnership between facultative microbes with hydrogenotrophic methanogens was previously reported.[44] The abundance of facultative *Anaerolinaceae__T78*[45] in test reactors was higher than in the control (3−9 vs 1%). Consistently, hydrogenotrophic *Methanobacterium* and *Methanoculleus*[46] were more abundant in test reactors (4−7 and 21−36%, respectively) than in the control (2 and 9%, respectively). As suggested in the literature,[47] *Anaerolinaceae__T78* might be involved in biodesulfurization as sulfide-oxidizing bacteria, as reasonably indicated by the highest abundance in the microaeration reactor. Compared to the control, the abundance of potential ARG hosts *Pseudomonas* and *Solibacillus* species[48,49] was lower in test reactors, which was consistent with lower ARG abundances in these reactors (discussed later). *Pseudomonas* is known to have sulfate-reducing capability.[50] Therefore, the results indicated that both

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microaeration and iron dosing suppressed sulfate-reducing activity. Despite this, total methane yields were not impacted and could provide insights into the possible functions of microbes enriched by microaeration and iron dosing.

The chord diagram in Figure 4c shows the relationship between various bacterial and archaeal genera present in the reactors. *Anaerolinaceae__T78*, *Caldicoprobacter*, *Clostridium*, and *Aminobacterium* were found to be closely associated with *Methanobacterium*, *Methanoculleus*, and *Methanosaeta*. Additionally, the acetogenic genus *vadinCA02* was found to have a strong connection to *Methanoculleus* and *Methanosaeta*. On the other hand, *Pseudomonas* and *Solibacillus* were only linked to each other and with other archaeal genera like *Methanosarcina* and *Methanomassiliicoccus*.

Antibiotic Resistance Genes. Among different tetracycline genes, *tet*W was the most abundant in all reactors with the highest level in the microaeration reactor (Figure 5a). Compared to the control, the abundances of *tet*M and *tet*X were lower in microaeration and iron-dosed reactors, whereas other *tet* genes (*tet*A, *tet*Q, *tet*W, and *tet*A) demonstrated higher or almost similar levels. Also, the total copies of *tet* genes in the microaeration and iron-dosed reactors were 7 and 10% lower than the control. The copies of sulfonamide ARGs (*sul*1 and *sul*2) were the most abundant ARGs in all reactors. Notably, sulfonamide is one of the frequently detected antibiotics in sludge and WWTPs.[51] The results indicated that the copies of *sul*1 in the microaeration and iron-dosed reactors were 67% lower compared to the control, while the abundance of *sul*2 was statistically significantly lower (p-value = <0.001) in the control. The copies of *erm*B genes in the control were higher (p-value = <0.001) than in both test reactors; however, *erm*C was more abundant (p-value = <0.001) in the microaeration reactor than in the control and iron-dosed reactors. It is worth noting that the abundance of various ARGs was within the range of absolute abundance of ARGs about 10^4 to 10^{10} copies per gram of sludge reported for digestate from full-scale sludge mesophilic AD in the literature.[52,53] Overall, the total copies of ARGs for the microaeration and iron-dosed reactors were 8.8×10^4 and 9.1×10^4 per gram of sludge which was significantly lower (p-value = <0.001) than the control (1.2×10^5 gene copies/g sludge) (Figure 5). Analogous to ARGs, MGEs (*intl*1 and *intl*2) also showed similar patterns, i.e., substantially higher levels (p-value = <0.001) in the control than in test reactors. Noteworthy, *intl*1 and *intl*2 can mediate DNA transmission by encoding specific proteins, hence inducing the HGT of ARGs.[54,55] The removal of *intl*2 was more prominent in both test reactors. However, between the two test reactors, the microaeration reactor had a slightly higher total abundance of integrons (*int*1 and *int*2). A few studies previously suggested that microaeration and iron dosing could improve the removal of ARGs.[13,56,57] Interestingly, the abundance of potential ARGs/MGE hosts, genera *Pseudomonas* and *Solibacillus*[48,49] were the most abundant in the control reactor (discussed earlier).

To further investigate the relationship between ARG abundance and microbial communities, a correlation analysis was conducted (Figure 6). A strong positive correlation was observed between the high abundance of ARG copies and the enrichment of *Solibacillus* (r = 0.98) and *Pseudomonas* (r = 1). On the other hand, all other genera showed a high negative correlation with the total ARG gene copies. For instance, *Aminobacterium* (r = −0.99), *Caldicoprobacter* (r = −97), *Cloacamonaceae_W22* (r = −0.94), *Clostridium* (r = −0.76), and *Anaerolinaceae__T78* (r = −0.7) showed a relatively weak correlation with ARG copies. Thus, the results indicated that potential changes in microbial communities due to microaeration and iron dosing could be the critical factor influencing the proliferation of ARGs/MGEs. However, caution should be exercised when interpreting such correlations, as the composition of bacterial communities often follows seasonal patterns.[58] Although the composition of ARGs does not typically show such seasonality, the absolute abundance of ARGs in incoming sludge may vary between winter and summer months.[59] Thus, the temporal patterns of microbial taxonomy and function may be asynchronous or uncoupled and warrant further investigation.

Significance of Results and Future Considerations. The results suggested that both methods could be effective for H_2S removal (up to 88%) from biogas. Both methods could provide additional benefits in terms of higher ARG/MGE removal than the control. However, depending on the end use of biogas, downstream purification processes may still be required to remove mercaptans, residual siloxane, and H_2S. For instance, siloxane concentrations below 0.03−30 mg/m^3 have been recommended for biogas utilization in combined heat and power (CHP) plants.[60,61] Overall, considering the technical performance observed in this study, there was no clear winner between the two methods. In addition to capital costs, the differences in electricity cost for microaeration operation and chemical cost may considerably impact process economics. Electricity prices may fluctuate considerably, and the costs of chemicals are subject to supply chain interruptions, as noticed in recent years. The iron dosing would also require the operation of a chemical dosing pump and paddle mixer, increasing the operating costs via electricity usage. Also, maintaining a continuous supply of chemicals (iron salt) can often be a concern for some facilities. Therefore, detailed capital costs and maintenance requirements/costs should be assessed along with the technical performance.

CONCLUSIONS

This study compared microaeration and iron dosing for in situ biogas desulfurization in AD. Up to 88% H_2S removal efficiencies were achieved with both methods. Moderate siloxane D4 removals (30−37%) were achieved with both methods, but they appeared to be ineffective for mercaptan removal. Despite no impact on methane production and sludge degradation efficiencies, both methods could alter microbial community composition by increasing the abundance of certain fermentative and facultative microbes and hydrogenotrophic methanogens compared with the control. The removal of most ARGs/MGEs was also enhanced in test reactors, which also corroborated with a decrease in potential ARG/MGE hosts. However, the transferability of these results to a full-scale application would require further site-specific assessment, pilot refinement, and economic evaluations. Furthermore, future research should prioritize long-term operations and delve into process design, control/monitoring, and maintenance requirements of these methods.

ASSOCIATED CONTENT

Supporting Information

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Schematic representation and photographs of the experimental set-up (Figure S1); methane and H_2S content in biogas measured during the pre-experimental phase (Figure S2); effluent TCOD, SCOD, TS, and VS concentration COD balance under different experimental phases (Figure S3); bacterial relative abundance at the phylum level in phase-3 (Figure S4); and α diversity metrics of the microbial community in different reactors in phase-3 (Table S1) (PDF)

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Notes

The authors declare no competing financial interest.

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